Show that $\mathbb{R}$ is a vector space over itself with respect to the usual "addition" and "scalar multiplication".

Sol^n: It is clear that $\forall\, a, b \in \mathbb{R} \Rightarrow a+b \in \mathbb{R}$

and $\forall\, a \in \mathbb{R}$, $\forall\, b \in \mathbb{R} \Rightarrow a.b = ab \in \mathbb{R}$.

It also has the following axioms:

(i) $\forall\, a, b \in \mathbb{R} \Rightarrow a+b = b+a$

(ii) $\forall\, a, b, c \in \mathbb{R} \Rightarrow (a+b)+c = a+(b+c)$

(iii) $\exists\, 0 \in \mathbb{R} \Rightarrow a+0 = 0+a = a \quad \forall\, a \in \mathbb{R}$

(iv) $\forall\, a \in \mathbb{R}$, $\exists\, -a \in \mathbb{R} \Rightarrow a+(-a) = (-a)+a = 0$

(v) $\forall\, a \in \mathbb{R}$ and $\forall\, b, c \in \mathbb{R} \Rightarrow a(b+c) = ab+ac$

(vi) $\forall\, a, b \in \mathbb{R}$ and $\forall\, c \in \mathbb{R} \Rightarrow (a+b)c = ac+bc$

(vii) $\forall\, a, b \in \mathbb{R}$ and $\forall\, c \in \mathbb{R} \Rightarrow (ab)c = a(bc)$

(viii) $1 \in \mathbb{R}$, $1 \neq 0$ and $\forall\, a \in \mathbb{R} \Rightarrow 1.a = a$

So, real number system is a vector space over itself.

Theorem: Let V be a vector space over a field $\mathbb{K}$. Then prove that,

(i) $\forall k \in \mathbb{K}$ and $0, u \in V \Rightarrow k0 = 0$

(ii) If $ku = 0 \Rightarrow k = 0$ or $u = 0$ where $k \in \mathbb{K}$ and $u \in V$.

Proof: (i) Given, $0 \in V \Rightarrow 0 + u = u + 0 = 0$, $\forall u \in V$

If $u = 0$ then we have $0 + 0 = 0$.— (i)

Now multiply both sides of (1) by k

$$k(0+0) = k0, \quad k \in \mathbb{K}$$

$$\Rightarrow k0 + k0 = k0 \quad \text{as } k(u+v) = ku + kv$$

$$\Rightarrow k0 + k0 = k0 + 0 \quad \text{as } u = u + 0$$

$$\Rightarrow k0 = 0$$

(ii) consider $ku = 0$ and $k \neq 0$

$$u = 1 \cdot u$$

$$= (k^{-1}k)u \quad [\because k \neq 0 \Rightarrow k^{-1} \in \mathbb{K} : k^{-1}k = 1]$$

$$= k^{-1}(ku) \quad [(kk^{-1})u = k(k^{-1}u)]$$

$$= k^{-1} \cdot 0$$

$$= 0$$

conversely, $ku = 0$ and $u \neq 0$

If $k \neq 0$ in this relation we get $u = 0$

This is a contradiction

$\therefore k = 0$ (proved)